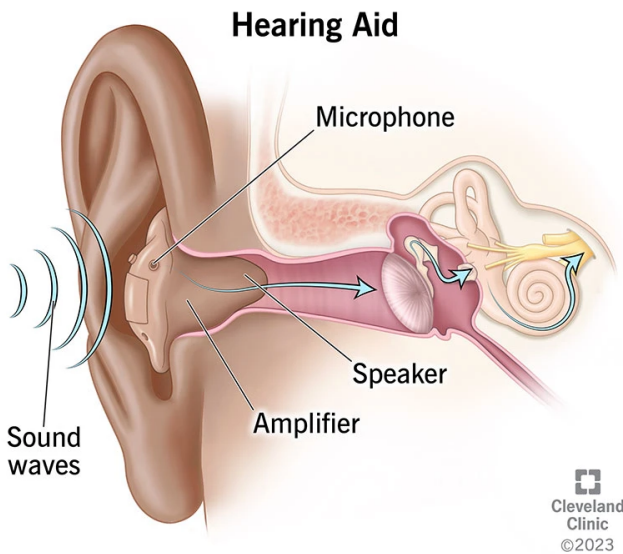


24<sup>th</sup> September 2026



Hearing aid whistle / Acoustic-feedback

Amped sound leaks out & is picked up by external mic again

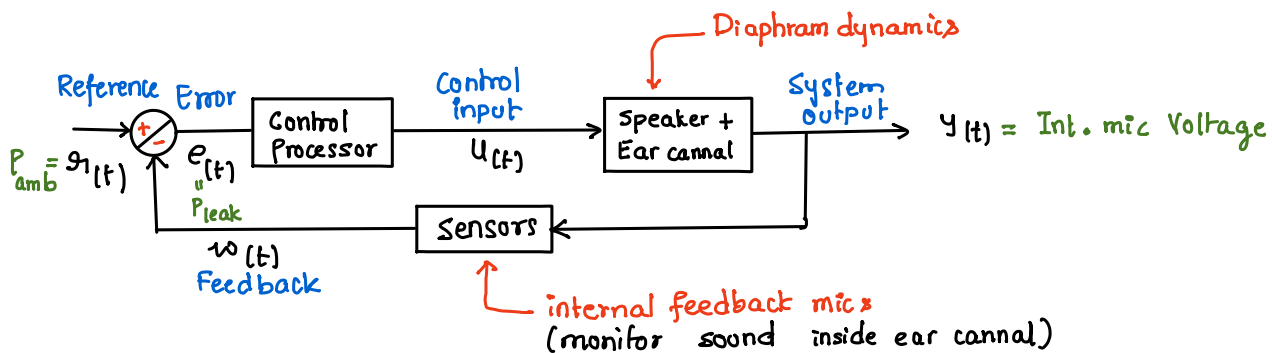
- Improper fit
- Wax build up
- Too much amp
- External obstruction
  - playing hand
  - hat
  - Hug etc. can still bounce back leaked sound into ext. mic

### TYPES OF HEARING AIDS

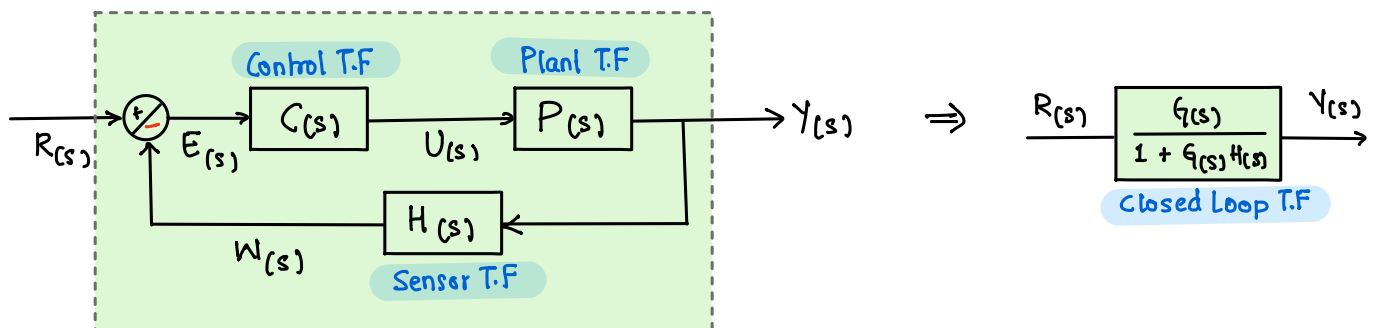


\* How can we eliminate this issue ?

\* Can we formulate this problem as a control problem ?

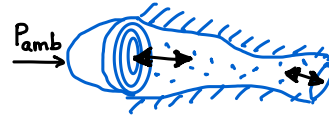


We want to model this as LTI:



## Lumped Model for air in a narrow vent:

$$m \ddot{y}(t) + b \dot{y}(t) + k y(t) = u(t)$$



$m$ : Effective acoustic - equivalent inertial mass of air in vent

$b$ : Viscous acoustic damping constant of air in vent

$k$ : Acoustic stiffness of the enclosed air volume

$$u(t) = A x(t) = A(y_{\text{ambient}} + y)$$

→ Acoustic feedback effect

A "better" approximation:

$$\mu(x,y) \frac{\partial^2 u}{\partial t^2} + c \frac{\partial u}{\partial t} - T(\nabla^2 u) = P(x,y) + \text{B.C. \& I.C.}$$

$u(x,y)$ : Diaphragm lateral displacement

$\mu(x,y)$ : Area density =  $\rho(x,y) \cdot h(x,y)$

$c$ : Damping factor

$T$ : Radial surface tension

$P(x,y,t)$ : Continuous incoming pressure waves

Different parameter values  
can model "Unhealthy  
ear"

Lumped model:

- 1.> **Device is OFF:**  $A = 0$  (no amplification) then  $y_{\text{internalmic}} = 0, \forall t$   
(Zero initial condition always)
- 2.> **Device is ON:**  $A > 1$  then acoustic feedback can occur  $u(t) = A(y_{\text{amb}} + y)$   
↓ the system dynamics (in terms of  $y_{\text{internalmic}} \rightarrow \infty$ )  
should exhibit instability
- 3.> **Can controller be designed for this?**  $u(t) = A y_{\text{externalmic}} + A(K_P E)$

\* Device OFF:

$$m \ddot{y} + b \dot{y} + k y = u(t) = 0$$

↓ for zero-initial conditions ⇒

$$y(t) = \dot{y}(t) = 0, \forall t$$

↓  $m, b, k$  all  $> 0$  for stable plant

Person will not hear  
anything

\* Device ON:

$$m \ddot{y} + b \dot{y} + k y = u(t) = A(y_{\text{ambient}} + y)$$

$$m \ddot{y} + b \dot{y} + (k - A) y = A y_{\text{amb}}$$

$$(ms^2 + bs + (k - A)) Y(s) = A R(s) \Rightarrow G(s) = \frac{Y(s)}{R(s)} = \frac{A}{(ms^2 + bs + k - A)}$$

$$\lambda = \frac{-b \pm \sqrt{b^2 - 4m(k-A)}}{2m} ; \quad * \text{ if } (A=k):$$

$$(m, b, k) > 0$$

$$\lambda = \frac{-b}{2m} \pm \frac{b}{2m} = \left(0 \text{ } 1 - \frac{b}{m}\right) \text{ stable}$$

$$* \text{ if } (A < k): (k-A) = c > 0$$

→ Overdamped & Critical:  $b^2 \geq 4ac$

$$\lambda = \frac{-b}{2m} \pm c_1, \quad \text{but } c_1 < \left(\frac{b}{2m}\right) \\ \therefore \text{ Stable}$$

→ Underdamped:  $b^2 < 4ac$

$$\lambda = \frac{-b}{2m} \pm j c_1, \quad \text{Stable with } -b/2m$$

But if  $(A > k) \Rightarrow$  Increasing signal amplification

-ve stiffness  $\rightarrow (k-A) = c < 0 \Rightarrow$  Unstable

↳ Represents the acoustic feedback  $\rightarrow$  But we desire higher amplification

Control design:

1. > Proportional Controller:  $m\ddot{y} + b\dot{y} + ky = u(t) = A(y_{amb} + y) - A(k_p y)$   
 $= Ay_{amb} + Ay(1 - k_p)$

$$\Rightarrow m\ddot{y} + b\dot{y} + (k - A(1 - k_p))y = Ay_{amb}$$

$$Y(s)(ms^2 + bs + k - A(1 - k_p)) = AR(s)$$

$$\text{if } (m, b, k) > 0 \\ \& \text{ for } (A > k)$$

$$CLTF = \frac{Y(s)}{R(s)} = \frac{A}{(ms^2 + bs + k - A(1 - k_p))}$$

$$\text{closed loop poles: } ms^2 + bs + k - A(1 - k_p) = 0$$

$$\lambda = \frac{-b}{2m} \pm \frac{1}{2m} \sqrt{b^2 - 4m[k - A(1 - k_p)]}$$

$$\therefore (m, b, k) > 0 \\ \& (A > k)$$

stability if:

$$k - A(1 - k_p) > 0$$

$$-A(1 - k_p) > -k$$

$$1 - k_p < (k/A)$$

$$1 - (k/A) < k_p \quad \because \left(\frac{k}{A}\right) < 1$$

$\Rightarrow$

$$1 - \left(\frac{k}{A}\right) < k_p < \infty //$$

Marginal if:

$$k_p = 1 - \left(\frac{k}{A}\right) //$$

Unstable if:

$$k_p < 1 - \left(\frac{k}{A}\right)$$

$$0 < k_p < 1 - \left(\frac{k}{A}\right) //$$

Recollect

★ Motivation to derivative controller:

for control augmented system

$$\left[ \lambda^2 + (2\tilde{\zeta}_n \tilde{\omega}_n) \lambda + \tilde{\omega}_n^2 = 0 \right]$$

$$\boxed{\tilde{\omega}_n^2 = k - A(1 - k_p)} \Rightarrow \left[ k_p = 1 - \left(\frac{\tilde{\omega}_n^2 - k}{A}\right) \right]$$

$$2\tilde{\zeta}_n \tilde{\omega}_n = b, \text{ fixing } \tilde{\omega}_n \text{ also fixes } \tilde{\zeta}_n$$

★ So derivative controller becomes a natural choice to have damping in our controller

\* if

$$\left. \begin{array}{l} m=1 \\ b=2 \\ k=3 \end{array} \right\} > 0$$

$$d \quad A=5 > k$$

then • marginal stability  $k_p = 1 - \frac{k}{A} = 1 - \frac{3}{5} = 0.4$   
/critical  $k_p$

$$\begin{aligned} \bullet \quad \text{CLTF} &= \frac{A}{(ms^2 + bs + [k - A(1 - k_p)])} = \frac{G(s)H(s)}{1 - G(s)H(s)} = \frac{G(s)}{1 - G(s)} \\ &= \frac{A}{(ms^2 + bs + k - A)} \\ &= \frac{A}{\left(1 + \frac{Ak_p}{(ms^2 + bs + k - A)}\right)} \end{aligned}$$

$$\text{OLTF} = G(s) = P(s)C(s) = \frac{Ak_p}{(ms^2 + bs + k - A)}$$